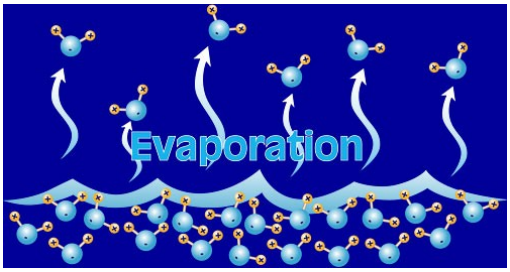
# Separation Techniques



A substance **that** dissolves in **a** liquid is said to **be** soluble in that liquid. The dissolved substance **forms** a solution. The soluble substance that dissolves **is** called the solute. The solvent is the substance that dissolves the solute. In a cup **of** coffee the solvent is water **and** the two solutes **are** coffee and sugar.

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Evaporation **can** be used to separate solutions such **as** salt dissolved in water. The solution is left **in** the Sun **to** allow the water to change **its** state to gas and leave **the** salt behind. With this method the solvent **is** lost to the atmosphere **and** the solute is collected. Evaporation works **on** mixtures.

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Distillation **is** similar to evaporation except **that** when the solvent is changed **to** a gas **it** is collected **in** a separate container **and** cooled back to **a** liquid. The result is both **the** solvent and solute can be collected again. Distillation works **on** mixtures.

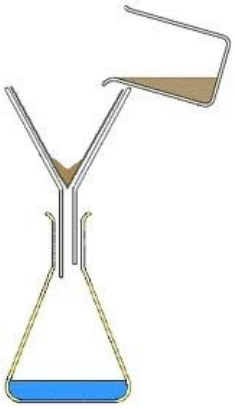
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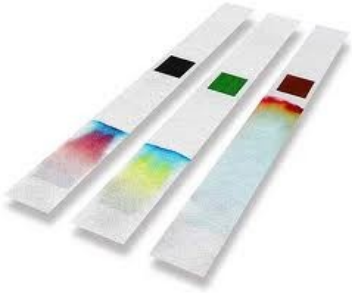
Decantation is **used** to separate **a** heavy insoluble solid **from** a liquid. The solid **is** allowed to sink **to** the bottom **of** the container and then the solvent is poured **off**, leaving the solid behind in **the** container. Sand **and** water can usually be separated **in** this way.

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Filtration is used **to** separate a fine insoluble solid **from** a liquid. The liquid **with** the solid suspended in **it** is poured through **a** filter paper which allows **the** solvent to pass through **and** catches the solid. Muddy river water can **be** filtered.



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Chromatography is a method **of** separating substances that **only** occur in small amounts in **a** mixture. The substance **in** a liquid form is allowed **to** pass through a special paper. The different molecules travel different distances through **the** paper and show **up** as coloured bands.

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Crystallisation is **a** technique used **to** separate a compound from a solution. Compounds **tend** to be more soluble **in** hot liquids than cold liquids. If a saturated hot solution **is** allowed to cool, the solute is **no** longer soluble in **the** solvent and forms crystals **of** pure compound. The crystals can **be** collected by filtration.

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